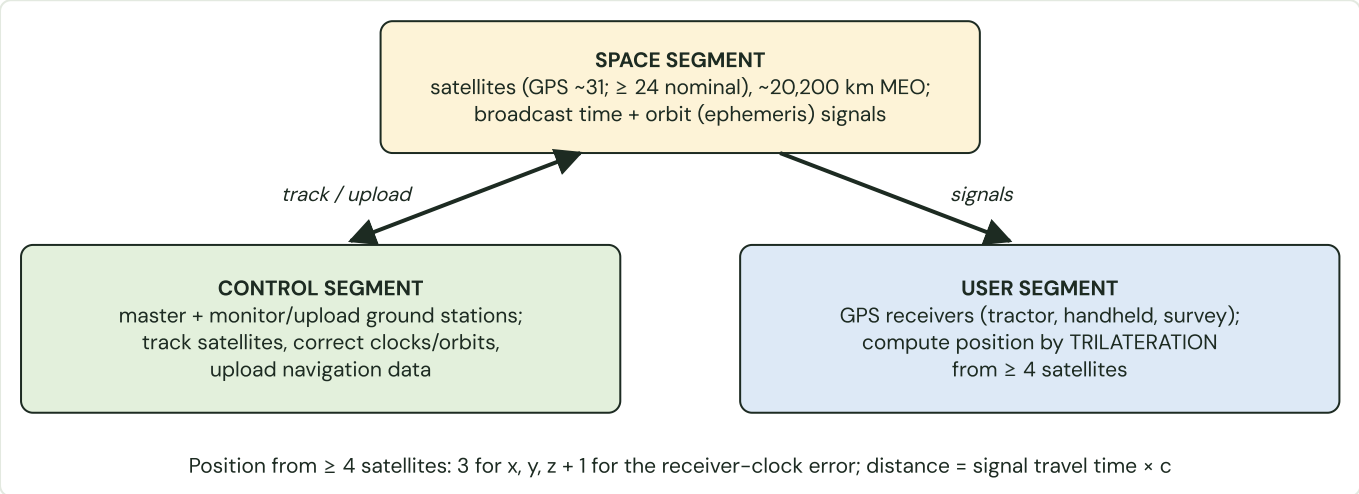
UNIT 10 · TOPIC 3(A)

GPS — The Three Segments, Positioning and Farm Uses

Syllabus topic: GPS basics: space, receiver and control segments; error and correction; function and usage

The three segments



GPS: the three segments (redrawn — the source figure had an icon printed over the satellite count)

Segment	What it is
SPACE	The satellites — nominally ≥ 24, currently ~31 — in ~20,200 km medium Earth orbit, each broadcasting coded time and orbital (ephemeris) signals
CONTROL	Ground network — master control station + monitor and upload stations worldwide — that tracks the satellites, corrects their clocks and orbits, and uploads updated navigation data
USER	GPS RECEIVERS (tractor-mounted, handheld, survey) that receive the signals and compute position

GPS = the Global Positioning System of the USA (NAVSTAR).

How a position is found — trilateration

1. The receiver measures the travel time of signals from several satellites.
2. Distance to each satellite = travel time × speed of light.
3. The position is where the range spheres intersect.

- At least **FOUR** satellites are needed — three to solve x, y, z and a fourth to correct the receiver's clock error.
- More satellites and good GEOMETRY (low DOP) give a better fix.

Functions and uses in farming

- Guidance and auto-steer.
- Parallel swathing — no overlaps or skips.
- Geo-referencing of samples and yield.

- Mapping.
- Variable-rate application.

Practice MCQs

Q9. GPS was developed by:

(a) India (b) USA
(c) Russia (d) China

Q10. The three GPS segments are space, control and:

(a) User (b) Weather
(c) Fuel (d) Radio

Q11. The GPS satellites constitute the:

(a) Control segment (b) Space segment
(c) User segment (d) Ground segment

Q12. Ground stations that track satellites and update their data form the:

(a) Space segment (b) Control segment
(c) User segment (d) Ionosphere

Q13. A GPS receiver on a tractor belongs to the:

(a) Space segment (b) Control segment
(c) User segment (d) Master station

Q14. GPS position is computed by:

(a) Triangulation of angles (b) Trilateration of distances
(c) Guesswork (d) Compass bearing

Q15. The minimum number of satellites for a 3-D position fix is:

(a) 1 (b) 2
(c) 3 (d) 4

Q16. The fourth satellite is needed to correct the:

(a) Orbit (b) Receiver clock error
(c) Weather (d) Antenna

Q17. GPS satellites orbit at an altitude of about:

(a) 400 km (b) 2,000 km
(c) 20,200 km (d) 36,000 km

Q18. Distance from a satellite is found from:

(a) Signal travel time × speed of light (b) Colour
(c) Voltage (d) Temperature

Q19. Satellite geometry quality is expressed by:

(a) DOP (Dilution of Precision) (b) BHN
(c) GF (d) pH

Q20. A LOWER DOP value means:

(a) Worse accuracy (b) Better positional accuracy
(c) No signal (d) More satellites failed

Q35. In farming, GPS is used for all EXCEPT:

(a) Auto-steer & parallel swathing (b) Geo-referencing samples & yield
(c) Variable-rate application (d) Measuring fuel calorific value

Answers

Q9 — (b) USA

Q10 — (a) User — *Three segments = space (satellites), control (ground stations), user (receivers).*

Q11 — (b) Space segment — *Space segment = satellites.*

Q12 — (b) Control segment — *Control segment = ground stations.*

Q13 — (c) User segment — *User segment = receivers.*

Q14 — (b) Trilateration of distances — *Trilateration from distances.*

Q15 — (d) 4 — ≥ *4 satellites — 3 for x, y, z and 1 for the receiver-clock error.*

Q16 — (b) Receiver clock error — *The 4th satellite resolves the receiver clock error.*

Q17 — (c) 20,200 km

Q18 — (a) Signal travel time × speed of light — *Distance = travel time × c.*

Q19 — (a) DOP (Dilution of Precision)

Q20 — (b) Better positional accuracy

Q35 — (d) Measuring fuel calorific value — *Fuel CV is measured by a calorimeter (Unit 9, Topic 6), not GPS — a cross-unit distractor.*

Sources: Srinivasan — Handbook of Precision Agriculture; NIIT/ICAR precision-agriculture modules; NovAtel — Introduction to GNSS; ISRO/PIB NavIC data (as given in the source notes, July 2026; operational status updated October 2026); NPTEL precision-agriculture lectures.